BIODEGRADABLE & ORGANIC WASTE ECONOMICS

Accesco Living

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Production & Processing Costs

Cost item	Collected figure	Basis
Labour cost	**₹780–₹940 per worker per day**	Collected
Worker productivity	**≈ 200–240 kg raw waste per worker per day**	Collected
Derived labour cost	**≈ ₹3.25–₹4.70 per kg raw waste**	Derived: ₹780 ÷ 240 kg to ₹940 ÷ 200 kg
Microbial inoculant	**₹0.20–₹0.24 per kg raw waste**	Collected
Curing / maturation	**₹0.50–₹1.50 per kg raw waste**	Collected
Segregation / processing	**Covered by labour cost above – not counted again**	Same activity: no double-counting
Packaging cost	**Depends on bag size and type**	Collected – see Yield & Total Cost

Yield & Total Cost

Yield (Bengaluru benchmark)

100 kg raw wet organic waste → **12–15 kg** finished compost

Calculation (derived)

Processing cost: ₹3.95–₹6.44 × 100 kg = **₹395–₹644 per 100 kg raw waste** (before packaging)

Cost per kg of **finished compost** (before packaging):
Best case: ₹395 ÷ 15 kg = **₹26.3**
Worst case: ₹644 ÷ 12 kg = **₹53.7**

Raw-waste cost and finished-compost cost differ because only 12–15% of the waste becomes compost.

Bag	Plain bag	Printed bag	Packaging per kg compost	Total cost per kg compost
1 kg	₹1.50–₹3.00	₹3.50–₹6.00	₹1.50–₹6.00	**₹27.8–₹59.7**
5 kg	₹4.00–₹7.00	₹8.00–₹12.00	₹0.80–₹2.40	**₹27.1–₹56.1**
25 kg	₹9.00–₹14.00	₹15.00–₹22.00	₹0.36–₹0.88	**₹26.7–₹54.6**

Actual total cost depends on the packaging format. Smaller bags add more packaging cost per kg; the main driver of cost is still processing cost and yield.

Selling Price & Revenue

Channel (Bengaluru)	Selling price	Price per kg	Revenue from 12–15 kg compost
Bulk: loose	₹2.05–₹3.50 /kg	₹2.05–₹3.50	**₹24.60–₹52.50**
Bulk: packed in 50 kg bags	₹5.50–₹8.40 /kg	₹5.50–₹8.40	**₹66.00–₹126.00**
Retail: 1 kg bag	₹25–₹65 per bag	₹25.0–₹65.0	**₹300–₹975**
Retail: 5 kg bag	₹115–₹299 per bag	₹23.0–₹59.8	**₹276–₹897**
Retail: 25 kg bag	₹300–₹450 per bag	₹12.0–₹18.0	**₹144–₹270**

Price per kg for retail = bag price ÷ bag size (derived). Revenue = lowest price × 12 kg to highest price × 15 kg, per 100 kg raw waste.

Transportation Cost

Tata Ace mini-truck (Bengaluru benchmark)	Collected figure
Capacity	**≈ 750–1,100 kg**
Base booking	**₹220–₹300**
Additional charge	**₹32–₹45 per km**

Transportation depends on **distance** and **vehicle utilization**.

The exact delivery distance is unknown, so no actual cost per kg is claimed. A small load is expensive per kg; filling the vehicle spreads the same trip cost over more kg.

Illustrative assumption: 10 km trip and 750 kg loaded capacity

Trip cost = ₹220–₹300 + (10 km × ₹32–₹45) = **₹540–₹750**, one-way (assumed).

Load carried (assumed)	Trip cost	Cost per kg of compost
750 kg (full vehicle)	₹540–₹750	**₹0.72–₹1.00**
25 kg (small load)	₹540–₹750	**₹21.60–₹30.00**

These are illustrative calculations, not measured figures, and are not included in the profit/loss analysis on the next slide.

Profit & Loss Analysis

Channel	Selling price ₹/kg	Cost ₹/kg compost*	Result
Bulk (loose / 50 kg bags)	₹2.05–₹8.40	₹26.3–₹53.7 (bag cost not collected)	**Loss in all cases**
Retail 1 kg	₹25–₹65	₹27.8–₹59.7	**Profit at higher prices; loss at low prices**
Retail 5 kg	₹23.0–₹59.8	₹27.1–₹56.1	**Profit at higher prices; loss at low prices**
Retail 25 kg	₹12.0–₹18.0	₹26.7–₹54.6	**Loss in all cases**

* Derived, per kg of finished compost, before transport. Bulk price is below calculated production cost, so bulk sales lose money under current cost and yield assumptions.

Examples per 100 kg raw waste (transport excluded)

Case	Assumptions	Cost	Revenue	Result
A. 1 kg, profit	Yield 15 kg; processing ₹395; plain bag ₹1.50/kg; sold at ₹65/kg	₹417.50	₹975.00	**Profit ₹557.50**
B. 1 kg, loss	Yield 12 kg; processing ₹644; printed bag ₹6.00/kg; sold at ₹25/kg	₹716.00	₹300.00	**Loss ₹416.00**
C. 5 kg, profit	Yield 15 kg; processing ₹395; plain bag ₹0.80/kg; sold at ₹59.8/kg (₹299 bag)	₹407.00	₹897.00	**Profit ₹490.00**
D. 25 kg, loss	Best-cost case: yield 15 kg; processing ₹395; plain bag ₹0.36/kg; sold at ₹18/kg	₹400.40	₹270.00	**Loss ₹130.40**

How to Minimise Cost & Maximise Profit

Reduce costs

- Improve worker productivity
- Reduce unnecessary labour handling
- Buy microbial inoculant in bulk
- Improve composting efficiency
- Reduce curing time where technically possible
- Use economical packaging
- Transport larger quantities per trip
- Avoid partially empty transport vehicles

Increase revenue

- Target higher-value retail customers
- Sell to nurseries, landscaping companies, farms and urban gardening businesses
- Negotiate better wholesale prices
- Charge waste generators a processing/service fee where applicable
- Treat waste-processing fees and compost sales as separate revenue streams

Bengaluru municipal opportunity: the default waste-management fee can be around ₹12/kg for applicable bulk generators, and in-situ processing can affect the fee/rebate structure. This is a **potential** opportunity, **not guaranteed revenue**.

Alternative Commercial Opportunities

Channel	Collected figure	Note
CBG / biogas	**≈ ₹70–₹105 per kg of finished gas**	Biogas route for wet waste
Animal feed	**≈ ₹8.50–₹12 per kg for some raw feed components; processed feed up to ≈ ₹20 per kg**	Only where the waste is clean and suitable
Waste-processing / user fees	**≈ ₹12 per kg default fee for applicable bulk generators (Bengaluru)**	Potential only; depends on generator type and in-situ rebate

Sources

#	Source	URL
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10	NITI for States – policy document	https://www.nitiforstates.gov.in/public-assets/Policy/policy_files/GNC491E000169.pdf?utm_source=chatgpt.com
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